

LEARNING PLAN – MANAGE DATABASE SYSTEMS

1. INSTITUTIONAL DETAILS

ITEM	DETAILS
Unit of Competence	Manage Computerized Database Systems
Unit Code	0612 451 08A
Name of Trainer	JAMES NDIRANGU
Competence Level	LEVEL 5
Institution	THE ABERDARE TECHNICAL COLLEGE
Number of Learners	2
Date of Preparation	06/08/2026
Skill or Job Task	Designing computerized databases, creating computerized databases, manipulating computerized databases, testing computerized databases and maintaining computerized databases.

2. BENCHMARK / CRITERIA TO BE USED

1. Design computerized database

- 1.1 Computerized database requirements are established as per user requirements and workplace procedures.*
- 1.2 Appropriate database models are selected according to database requirements.*
- 1.3 Conceptual database design is developed according to user requirements.*
- 1.4 Logical database design is developed according to the conceptual design.*
- 1.5 Physical database design is developed according to the logical design and user requirements.*

2. Create computerized database

- 2.1 Database objects are created according to the database design.*
- 2.2 Tables and fields are created according to database requirements.*
- 2.3 Tables are linked according to identified relationships.*

- 2.4 Queries, forms and reports are created according to user requirements.*
- 2.5 Integrity constraints are implemented according to database requirements.*
- 2.6 Database indexing is implemented according to database requirements.*

3. Manipulate computerized database

- 3.1 Database is populated according to database requirements.*
- 3.2 Database operations are performed according to user requirements.*
- 3.3 Database queries are developed using appropriate SQL commands.*
- 3.4 Database views are created and managed according to requirements.*
- 3.5 Database reports are generated, edited, formatted and printed according to user requirements.*

4. Test computerized database

- 4.1 Database test plan is developed according to database requirements.*
- 4.2 Appropriate database test data is prepared according to the test plan.*
- 4.3 Database testing is performed using appropriate test data.*
- 4.4 Database results are validated against expected results.*
- 4.5 Database test report is prepared according to workplace requirements.*

5. Maintain computerized database

- 5.1 Database optimization techniques are applied according to performance requirements.*
- 5.2 Computerized database maintenance schedule is prepared according to workplace procedures.*
- 5.3 Computerized database is monitored according to the maintenance schedule.*
- 5.4 Database backup and recovery procedures are performed according to workplace requirements.*
- 5.5 Computerized database maintenance report is prepared according to workplace procedures.*

3. LEARNING PLAN

LEARNING OUTCOME 1: DESIGN COMPUTERIZED DATABASE

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources and References	Assessment	Reflections & Dates
1	1. Design computerized database	Session 1: Introduction to DBMS and Database Requirements • Introduction to DBMS • Definition of terms • Functions of database systems • Database applications • Types of DBMS • Major components of DBMS environment	By the end of the session, trainees should be able to: 1. Define database and DBMS terminology. 2. Explain the functions of database systems. 3. Identify database applications. 4. Classify different types of DBMS. 5. Identify the major components of a DBMS environment.	1. Introduces DBMS concepts and poses questions on basic database terminology. 2. Explains database terms and demonstrates their application. 3. Demonstrates common database applications and asks trainees to identify them. 4. Explains different types of DBMS and asks trainees to classify examples. 5. Demonstrates major DBMS components and asks trainees to identify them.	1. Listen to the DBMS introduction and answer questions on database terminology. 2. Define database terms and give relevant examples. 3. Identify database applications demonstrated by the trainer. 4. Classify DBMS examples according to their types. 5. Identify the major DBMS components demonstrated by the trainer.	Resources: Computers, projector, DBMS software, Notes, Curriculum, Occupational standards. References: Elmasri, R., & Navathe, S. B. Fundamentals of Database Systems; Silberschatz, A., Korth, H. F., & Sudarshan, S. Database System Concepts.	Oral questions, terminology exercise, class discussion	Date: _____ Reflection: _____
1	1. Design	Session 2:	By the end of the	1. Explains	1. Describe	Resources:	Requirement	Date:

	computerized database	Database Architecture and Requirements Gathering • Database architecture • Requirements gathering techniques • Requirement elicitation • Database assessment requirement report	session, trainees should be able to: 1. Describe database architecture. 2. Identify requirements gathering techniques. 3. Apply requirement elicitation techniques. 4. Analyse user requirements. 5. Prepare a database assessment requirement report.	database architectures and poses questions on their differences. 2. Introduces interviews, questionnaires and observation and asks trainees to identify each technique. 3. Demonstrates requirement elicitation and guides trainees through a sample exercise. 4. Provides user requirements and guides trainees in analysing them. 5. Demonstrates report preparation and guides trainees in preparing a requirement report.	database architectures and answer questions on their differences. 2. Identify interviews, questionnaires and observation techniques. 3. Apply requirement elicitation techniques during the sample exercise. 4. Analyse the provided user requirements. 5. Prepare a database assessment requirement report.	Computers, projector, sample questionnaires, interview forms, report templates. References: Connolly, T., & Begg, C. Database Systems; Kendall, K. E., & Kendall, J. E. Systems Analysis and Design.	elicitation exercise, requirement report	_____ Reflection: _____
2	1. Design computerized database	Session 3: Database Models • Relational model • Hierarchical model	By the end of the session, trainees should be able to: 1. Identify different database	1. Introduces database models and poses identification questions.	1. Identify the database models and answer identification questions.	Resources: Database diagrams, computers, projector, DBMS	Model comparison exercise, oral questions, case study	Date: _____ Reflection: _____

		• Network model • Object-oriented model • Dimensional model	models. 2. Explain the characteristics of each model. 3. Differentiate database models. 4. Analyse suitable models for given applications. 5. Select an appropriate database model for a given requirement.	2. Explains characteristics of each model and asks trainees to describe them. 3. Compares database models and asks trainees to differentiate them. 4. Provides application scenarios and guides trainees in analysing them. 5. Presents database requirements and guides trainees in selecting suitable models.	2. Describe the characteristics of each model. 3. Differentiate the database models using their characteristics. 4. Analyse the application scenarios provided. 5. Select suitable database models for the given requirements.	software. References: Date, C. J. An Introduction to Database Systems; Harrington, J. L. Relational Database Design.		
2	1. Design computerized database	Session 4: Conceptual Database Design and ERDs • Conceptual design • ERDs • Types of entities and notations • Types of attributes and notations	By the end of the session, trainees should be able to: 1. Explain conceptual database design. 2. Identify entities and attributes. 3. Differentiate relationship types. 4. Construct an ERD.	1. Explains conceptual database design and poses questions on its purpose. 2. Demonstrates entities and attributes and asks trainees to identify them. 3. Explains	1. Explain conceptual database design and answer questions on its purpose. 2. Identify entities and attributes from the examples provided. 3. Differentiate relationship types	Resources: Computers, draw.io, ERD templates, projector. References: Chen, P. P. The Entity-Relationship Model; Coronel, C., & Morris, S.	ERD drawing practical	Date: _____ Reflection: _____

		Types of relationships and notations	5. Evaluate an ERD against user requirements.	relationship types and asks trainees to differentiate them. 4. Demonstrates ERD construction and guides trainees in drawing ERDs. 5. Provides sample ERDs and guides trainees in evaluating them.	using the given examples. 4. Construct ERDs using the demonstrated notation. 5. Evaluate sample ERDs against the stated requirements.	Database Systems.		
3	1. Design computerized database	Session 5: Relational and Logical Database Design - Relational model - Logical design - Defining tables - Keys - Normalization	By the end of the session, trainees should be able to: - Explain relational database design. - Identify tables and keys. - Apply normalization principles. - Analyse unnormalized data. - Develop a logical database design.	- Explains relational database design and poses questions on its principles. - Demonstrates tables and keys and asks trainees to identify them. - Demonstrates normalization and guides trainees through normalization exercises. - Provides unnormalized data and guides trainees in	- Explain relational database design and answer questions on its principles. - Identify tables and keys from the examples provided. - Apply normalization principles to the given data. - Analyse the unnormalized data and identify problems. - Develop a logical database	Resources: Computers, DBMS, normalization worksheets, ERD diagrams. References: Codd, E. F. A Relational Model of Data; Hernandez, M. J. Database Design for Mere Mortals.	Normalization exercise, logical design practical	Date: _____ Reflection: _____

				analysing it. 5. Demonstrates conversion of an ERD into tables and guides logical design development.	design from the ERD.			
3	1. Design computerized database	Session 6: Physical Database Design • Selection of data types • Indexing • Constraints	By the end of the session, trainees should be able to: 1. Identify appropriate data types. 2. Select suitable data types for fields. 3. Explain database indexing. 4. Apply database constraints. 5. Develop a physical database design.	1. Introduces database data types and poses selection questions. 2. Demonstrates data type selection and asks trainees to select appropriate types. 3. Explains database indexing and demonstrates index creation. 4. Demonstrates database constraints and guides trainees in applying them. 5. Provides a database design task and guides trainees in developing the physical design.	1. Identify appropriate data types and answer selection questions. 2. Select suitable data types for the given fields. 3. Explain indexing and create indexes as demonstrated. 4. Apply the appropriate database constraints. 5. Develop a physical database design from the logical design.	Resources: Computers, DBMS software, database design worksheets. References: Lightstone, S., et al. Physical Database Design; Elmasri, R., & Navathe, S. B. Fundamentals of Database Systems.	Physical design practical	Date: _____ Reflection: _____
4	1. Design	Session 7:	By the end of the	1. Provides a	1. Analyse the	Resources:	CAT 1:	Date:

	computerized database	Integrated Database Design • Requirements • Database model • Conceptual design • ERD • Logical design • Physical design	session, trainees should be able to: 1. Analyse database requirements. 2. Evaluate alternative database models. 3. Develop an ERD. 4. Create logical tables and keys. 5. Create a physical database design. 6. Evaluate the complete design against user requirements.	database case study and guides trainees in analysing requirements. 2. Presents alternative database models and asks trainees to evaluate them. 3. Demonstrates ERD development and guides trainees in developing ERDs. 4. Demonstrates conversion of ERDs into logical tables and guides trainees in creating tables and keys. 5. Demonstrates physical design development and guides trainees in creating the design. 6. Reviews completed designs and asks trainees to evaluate them against	database case study and identify the requirements. 2. Evaluate the alternative database models and select the appropriate model. 3. Develop an ERD from the identified requirements. 4. Create logical tables and keys from the ERD. 5. Create the physical database design. 6. Evaluate the completed design against the user requirements.	Computers, draw.io, DBMS, case studies. References: TVET CDACC ICT Level 5 Curriculum Standard; Connolly, T., & Begg, C. Database Systems.	Complete computerized database design practical	_____ Reflection: _____
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				requirements.				
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LEARNING OUTCOME 2: CREATE COMPUTERIZED DATABASE

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources and References	Assessment	Reflections & Dates
5	2. Create computerized database	Session 8: Database Objects, Fields and Data Types • Tables, Forms, Queries, Reports, Macros • Fields, Records • Data types, Field properties, Keys	By the end of the session, trainees should be able to: 1. Identify database objects. 2. Explain the purpose of each database object. 3. Differentiate fields and records. 4. Select appropriate data types. 5. Apply field properties and keys.	1. Introduces database objects and asks trainees to identify them. 2. Explains the purpose of each object and poses questions on their uses. 3. Demonstrates fields and records and asks trainees to differentiate them. 4. Demonstrates data type selection and guides trainees in selecting types.	1. Identify database objects and answer identification questions. 2. Explain the purpose of each database object. 3. Differentiate fields from records using the examples provided. 4. Select appropriate data types for given fields. 5. Apply field properties and keys as	Resources: Computers, MS Access/DBMS software, projector. References: Pratt, P. J., & Last, M. Z. Concepts of Database Management; Microsoft Office Documentation.	Database object identification, practical exercise	Date: _____ Reflection: _____

				5. Demonstrates field properties and keys and guides trainees in applying them.	demonstrated.			
5	2. Create computerized database	Session 9: Creation and Linking of Tables • Creation of tables • Primary keys • Foreign keys • Linking tables • Relationships	By the end of the session, trainees should be able to: 1. Create database tables. 2. Define primary keys. 3. Create foreign keys. 4. Link tables. 5. Evaluate table relationships.	1. Demonstrates table creation and guides trainees in creating tables. 2. Demonstrates primary key creation and asks trainees to define primary keys. 3. Demonstrates foreign key creation and guides trainees in creating foreign keys. 4. Demonstrates table linking and guides trainees in linking tables. 5. Provides relationship examples and asks trainees to evaluate them.	1. Create tables following the demonstrated procedure. 2. Define primary keys for the created tables. 3. Create appropriate foreign keys. 4. Link tables using the identified relationships. 5. Evaluate the table relationships and correct errors.	Resources: Computers, DBMS, sample datasets. References: Harrington, J. L. Relational Database Design; Coronel, C., & Morris, S. Database Systems.	Table creation and linking practical	Date: _____ Reflection: _____
6	2. Create computerized database	Session 10: Database Development – Queries, Forms	By the end of the session, trainees should be able to: 1. Explain the	1. Explains queries, forms and reports and poses questions on their	1. Explain the purposes of queries, forms and reports and	Resources: Computers, DBMS, sample database.	Queries, forms and reports practical	Date: _____ Reflection: _____

		and Reports • Creation of queries • Creation of forms • Creation of reports	purpose of queries, forms and reports. 2. Create database queries. 3. Create database forms. 4. Create database reports. 5. Evaluate database objects against user requirements.	purposes. 2. Demonstrates query creation and guides trainees in creating queries. 3. Demonstrates form creation and guides trainees in designing forms. 4. Demonstrates report creation and guides trainees in generating reports. 5. Provides user requirements and asks trainees to evaluate their objects.	answer questions. 2. Create queries according to the demonstrated procedure. 3. Create forms according to the given requirements. 4. Create reports using the sample database. 5. Evaluate the completed objects against user requirements.	References: Alexander, M., & Kusleika, R. Access 2019 Bible; Silberschatz, A., et al. Database System Concepts.		
6	2. Create computerized database	Session 11: Integrity Constraints and Database Indexing • Nulls • Entity integrity • Referential integrity • General constraints • Primary indexing • Secondary indexing	By the end of the session, trainees should be able to: 1. Explain database integrity. 2. Identify types of integrity constraints. 3. Apply entity and referential integrity. 4. Create primary and secondary indexes.	1. Explains database integrity and poses questions on its importance. 2. Demonstrates integrity constraints and asks trainees to identify them. 3. Demonstrates entity and referential integrity and	1. Explain database integrity and answer questions on its importance. 2. Identify the different integrity constraints demonstrated. 3. Apply entity and referential integrity to the database. 4. Create primary	Resources: Computers, DBMS, sample database. References: Date, C. J. Database in Depth; Elmasri, R., & Navathe, S. B. Fundamentals of Database Systems.	Integrity and indexing practical	Date: _____ Reflection: _____

			5. Evaluate database integrity.	guides trainees in applying them. 4. Demonstrates primary and secondary indexing and guides trainees in creating indexes. 5. Provides integrity scenarios and asks trainees to evaluate them.	and secondary indexes. 5. Evaluate database integrity and identify violations.			
7	2. Create computerized database	Session 12: Data Relationships • One-to-one • One-to-many • Many-to-one • Many-to-many relationships	By the end of the session, trainees should be able to: 1. Identify database relationships. 2. Explain relationship types. 3. Differentiate one-to-one, one-to-many, many-to-one and many-to-many relationships. 4. Apply relationships when linking tables. 5. Evaluate relationships for correctness.	1. Introduces database relationships and asks trainees to identify examples. 2. Explains relationship types and poses questions about their characteristics. 3. Compares relationship types and asks trainees to differentiate them. 4. Demonstrates relationship implementation and guides trainees in linking	1. Identify database relationships and give relevant examples. 2. Explain the characteristics of each relationship type. 3. Differentiate the relationship types using the given scenarios. 4. Apply relationships when linking database tables. 5. Evaluate the relationships and correct errors.	Resources: Computers, DBMS, ERDs, sample databases. References: Hernandez, M. J. Database Design for Mere Mortals; Connolly, T., & Begg, C. Database Systems.	Relationship practical	Date: _____ Reflection: _____

				tables. 5. Provides relationship scenarios and asks trainees to evaluate them.				
7	2. Create computerized database	Session 13: Integrated Database Creation • Tables, Relationships • Queries, Forms, Reports • Constraints, Indexing	By the end of the session, trainees should be able to: 1. Create database tables. 2. Link database tables. 3. Create queries, forms and reports. 4. Apply integrity constraints. 5. Implement database indexing. 6. Evaluate the completed database.	1. Demonstrates complete database creation and guides trainees in creating tables. 2. Demonstrates relationships and guides trainees in linking tables. 3. Demonstrates queries, forms and reports and guides trainees in creating them. 4. Demonstrates integrity constraints and guides trainees in applying them. 5. Demonstrates indexing and guides trainees in implementing indexes. 6. Reviews completed databases and asks	1. Create database tables from the provided design. 2. Link tables using appropriate relationships. 3. Create queries, forms and reports. 4. Apply appropriate integrity constraints. 5. Implement suitable database indexes. 6. Evaluate the completed database against the requirements.	Resources: Computers, DBMS, database design, sample data. References: TVET CDACC Assessment Guidelines; Elmasri, R., & Navathe, S. B. Fundamentals of Database Systems.	CAT 2 + Practical Assessment 1: Database Creation	Date: _____ Reflection: _____

				trainees to evaluate them.				
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LEARNING OUTCOME 3: MANIPULATE COMPUTERIZED DATABASE

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources and References	Assessment	Reflections & Dates
8	3. Manipulate computerized database	Session 14: Database Population and Operations • Database population • INSERT, SELECT, UPDATE, DELETE	By the end of the session, trainees should be able to: 1. Explain database population. 2. Insert data into a database. 3. Select required records. 4. Update existing records. 5. Delete records correctly. 6. Evaluate database operations.	1. Explains database population and poses questions on its purpose. 2. Demonstrates INSERT operations and guides trainees in inserting data. 3. Demonstrates SELECT operations and guides trainees in retrieving records. 4. Demonstrates UPDATE operations and guides trainees in modifying records. 5. Demonstrates DELETE operations and guides trainees in deleting records safely. 6. Provides database operations and	1. Explain database population and answer questions on its purpose. 2. Insert data into the database as demonstrated. 3. Select required records from the database. 4. Update existing database records. 5. Delete specified records correctly. 6. Evaluate the results of the database operations.	Resources: Computers, DBMS, SQL environment, sample database. References: Beaulieu, A. Learning SQL; DuBois, P. MySQL.	Database manipulation practical	Date: _____ Reflection: _____

				asks trainees to evaluate the results.				
8	3. Manipulate computerized database	Session 15: SQL and Database Querying • Introduction to SQL • Writing SQL commands • SQL views	By the end of the session, trainees should be able to: 1. Define SQL. 2. Explain common SQL commands. 3. Write basic SQL commands. 4. Apply SQL to retrieve and manipulate data. 5. Create SQL views.	1. Introduces SQL and poses questions about its purpose. 2. Explains common SQL commands and asks trainees to identify their uses. 3. Demonstrates SQL syntax and guides trainees in writing commands. 4. Demonstrates data retrieval and manipulation and guides trainees in executing SQL. 5. Demonstrates SQL view creation and guides trainees in creating views.	1. Define SQL and answer questions about its purpose. 2. Identify common SQL commands and state their uses. 3. Write SQL commands using the demonstrated syntax. 4. Apply SQL commands to retrieve and manipulate data. 5. Create SQL views according to the given requirements.	Resources: Computers, MySQL/SQL Server/Access, SQL worksheets. References: Forta, B. Sams Teach Yourself SQL in 10 Minutes; Viescas, J. L. SQL Queries for Mere Mortals.	SQL practical exercise	Date: _____ Reflection: _____
9	3. Manipulate computerized database	Session 16: Database Report Generation • Generate, Edit, Format report • Apply header/footer	By the end of the session, trainees should be able to: 1. Explain database report generation. 2. Generate	1. Explains report generation and poses questions on its purpose. 2. Demonstrates report generation and guides	1. Explain report generation and answer questions on its purpose. 2. Generate database reports from the sample	Resources: Computers, DBMS, printer, report templates. References: Microsoft Access	Report generation practical	Date: _____ Reflection: _____

		• Print report	database reports. 3. Edit database reports. 4. Format database reports. 5. Apply headers and footers. 6. Produce professional printed reports.	trainees in generating reports. 3. Demonstrates report editing and guides trainees in editing reports. 4. Demonstrates report formatting and guides trainees in formatting reports. 5. Demonstrates headers and footers and guides trainees in applying them. 6. Demonstrates report printing and guides trainees in producing final reports.	database. 3. Edit the generated reports. 4. Format reports according to the requirements. 5. Apply appropriate headers and footers. 6. Produce and print professional reports.	Report Design Guides; Coronel, C., & Morris, S. Database Systems.		
9	3. Manipulate computerized database	Session 17: Integrated Database Manipulation • Data population & operations • SQL, Views, Reports	By the end of the session, trainees should be able to: 1. Apply database operations. 2. Write SQL queries. 3. Create database views. 4. Generate and format database reports.	1. Provides a practical scenario and guides trainees in applying database operations. 2. Provides SQL tasks and guides trainees in writing queries. 3. Demonstrates view creation and	1. Apply database operations to the practical scenario. 2. Write and execute SQL queries. 3. Create database views from the given requirements. 4. Generate and format database	Resources: Computers, DBMS, SQL, sample database, printer. References: Beaulieu, A. Learning SQL; TVET CDACC Assessment Standards.	CAT 3: Database Manipulation Practical	Date: _____ Reflection: _____

			5. Analyse query results. 6. Evaluate database manipulation activities.	guides trainees in creating views. 4. Demonstrates report generation and guides trainees in generating and formatting reports. 5. Provides query results and asks trainees to analyse them. 6. Reviews the completed activities and asks trainees to evaluate their work.	reports. 5. Analyse the query results and identify findings. 6. Evaluate the completed database manipulation activities.			
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LEARNING OUTCOME 4: TEST COMPUTERIZED DATABASE

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources and References	Assessment	Reflections & Dates
10	4. Test computerized database	Session 18: Database Test Planning and Test Data • Importance & types of DB testing • Structural, Functional, Non-functional • Load, Stress, Stored procedures • Test data preparation	By the end of the session, trainees should be able to: 1. Explain the importance of database testing. 2. Identify database testing types. 3. Differentiate structural, functional and non-functional testing. 4. Analyse database testing requirements. 5. Prepare appropriate test data. 6. Develop a database test plan.	1. Explains the importance of database testing and poses questions on its purpose. 2. Introduces database testing types and asks trainees to identify them. 3. Compares structural, functional and non-functional testing and asks trainees to differentiate them. 4. Provides database requirements and guides trainees in analysing testing requirements. 5. Demonstrates test-data preparation and guides trainees in	1. Explain the importance of database testing and answer questions. 2. Identify the different database testing types. 3. Differentiate structural, functional and non-functional testing. 4. Analyse the database testing requirements provided. 5. Prepare appropriate test data. 6. Develop a database test plan.	Resources: Computers, DBMS, test templates, sample databases. References: Ambler, S. W., & Sadalage, P. J. Refactoring Databases; Myers, G. J. The Art of Software Testing.	Test plan and test-data preparation exercise	Date: _____ Reflection: _____

				preparing test data. 6. Demonstrates test-plan development and guides trainees in developing a plan.				
10	4. Test computerized database	Session 19: Performing Database Testing and Validation • Database testing using test data • Result validation • Structural & functional testing • White-box, Black-box, Load, Stress	By the end of the session, trainees should be able to: 1. Apply database testing procedures. 2. Perform structural testing. 3. Perform functional testing. 4. Perform non-functional testing. 5. Validate database results. 6. Evaluate database performance against expected results.	1. Demonstrates database testing procedures and guides trainees in applying them. 2. Demonstrates structural testing and guides trainees in testing database structures. 3. Demonstrates functional testing and guides trainees in testing functions. 4. Demonstrates non-functional testing and guides trainees in carrying out appropriate tests. 5. Provides expected and actual results and guides trainees in validating them.	1. Apply database testing procedures using the provided test cases. 2. Perform structural testing on the database. 3. Perform functional testing on database functions. 4. Perform appropriate non-functional tests. 5. Validate actual results against expected results. 6. Evaluate database performance and identify defects.	Resources: Computers, DBMS, testing tools, test data, test cases. References: Dusting, G. Database Testing Methodologies; IEEE Standard for Software Test Documentation.	Database testing practical	Date: _____ Reflection: _____

				6. Provides performance results and asks trainees to evaluate database performance.				
10	4. Test computerized database	Session 20: Database Test Report • Importance of test reports • Test report preparation • Test results & defects identified • Analysis & recommendations	By the end of the session, trainees should be able to: 1. Explain the importance of test reports. 2. Organize test results. 3. Analyse database test findings. 4. Prepare a database test report. 5. Present test findings. 6. Recommend appropriate corrective measures.	1. Explains the importance of test reports and poses questions on their purpose. 2. Demonstrates organization of test results and guides trainees in organizing them. 3. Provides test findings and guides trainees in analysing them. 4. Demonstrates test report preparation and guides trainees in preparing reports. 5. Demonstrates professional presentation of findings and guides trainees in presenting their findings. 6. Discusses	1. Explain the importance of test reports and answer questions. 2. Organize the provided test results. 3. Analyse the database test findings. 4. Prepare a database test report. 5. Present the test findings professionally. 6. Recommend appropriate corrective measures.	Resources: Computers, report templates, test results, projector. References: Software Engineering Institute (SEI) Technical Reporting Guidelines; TVET CDACC Framework.	CAT 4 / Database Testing and Test Report Practical	Date: _____ Reflection: _____

				identified defects and asks trainees to recommend corrective measures.				
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LEARNING OUTCOME 5: MAINTAIN COMPUTERIZED DATABASE

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources and References	Assessment	Reflections & Dates
11	5. Maintain computerized database	Session 21: Database Optimization • Introduction to optimization • Query optimization • Indexing strategies • Partitioning & storage management	By the end of the session, trainees should be able to: 1. Explain database optimization. 2. Identify factors affecting database performance. 3. Analyse poorly performing queries. 4. Apply query optimization techniques. 5. Apply	1. Explains database optimization and poses questions on its importance. 2. Identifies performance factors and asks trainees to identify them in sample databases. 3. Demonstrates poorly performing queries and guides trainees in analysing them.	1. Explain database optimization and answer questions on its importance. 2. Identify factors affecting database performance. 3. Analyse poorly performing queries. 4. Apply query optimization techniques to improve performance.	Resources: Computers, DBMS, sample databases, SQL tools. References: Tow, D. SQL Tuning; Lightstone, S., et al. Physical Database Design.	Query optimization practical	Date: _____ Reflection: _____

			appropriate indexing strategies. 6. Evaluate partitioning and storage management strategies.	4. Demonstrates query optimization techniques and guides trainees in applying them. 5. Demonstrates indexing strategies and guides trainees in applying them. 6. Explains partitioning and storage management and asks trainees to evaluate the strategies.	5. Apply suitable indexing strategies. 6. Evaluate partitioning and storage management strategies.			
11	5. Maintain computerized database	Session 22: Database Maintenance Schedule and Backup & Recovery • Maintenance techniques • Maintenance schedule preparation • Key schedule elements • Database backup & recovery	By the end of the session, trainees should be able to: 1. Identify database maintenance techniques. 2. Explain the importance of maintenance schedules. 3. Prepare a database maintenance schedule.	1. Explains preventive, corrective and adaptive maintenance and asks trainees to identify each type. 2. Explains the importance of maintenance schedules and poses questions on their purpose. 3. Demonstrates	1. Identify preventive, corrective and adaptive maintenance techniques. 2. Explain the importance of maintenance schedules and answer questions. 3. Prepare a database maintenance schedule.	Resources: Computers, DBMS, backup storage, maintenance templates. References: Mullin, M. Database Administration; Silberschatz, A., et al. Database System Concepts.	Maintenance schedule and backup/recovery practical	Date: _____ Reflection: _____

			4. Identify key maintenance activities. 5. Perform database backup. 6. Apply database recovery procedures.	maintenance schedule preparation and guides trainees in preparing schedules. 4. Identifies key maintenance activities and asks trainees to identify them from scenarios. 5. Demonstrates database backup and guides trainees in performing backups. 6. Demonstrates recovery procedures and guides trainees in recovering a database.	4. Identify key maintenance activities from the given scenarios. 5. Perform database backup using the demonstrated procedure. 6. Apply database recovery procedures and verify the recovered database.			
12	5. Maintain computerized database	Session 23: Computerized Database Monitoring • Database monitoring tools • Resource utilization • Query	By the end of the session, trainees should be able to: 1. Identify database monitoring tools. 2. Explain database monitoring	1. Introduces database monitoring tools and asks trainees to identify them. 2. Explains monitoring requirements and poses questions	1. Identify database monitoring tools and answer identification questions. 2. Explain database monitoring	Resources: Computers, DBMS monitoring tools, sample database, monitoring software. References:	Database monitoring practical	Date: _____ Reflection: _____

		performance • Transaction log & connection monitoring • Alerting systems	requirements. 3. Monitor resource utilization. 4. Analyse query performance. 5. Monitor transaction logs and connections. 6. Configure appropriate alerting mechanisms.	on their importance. 3. Demonstrates resource monitoring and guides trainees in monitoring resource utilization. 4. Demonstrates query performance monitoring and guides trainees in analysing performance. 5. Demonstrates transaction log and connection monitoring and guides trainees in monitoring them. 6. Demonstrates alert configuration and guides trainees in configuring alerts.	requirements and answer questions on their importance. 3. Monitor database resource utilization using the available tools. 4. Analyse query performance and identify performance problems. 5. Monitor transaction logs and database connections. 6. Configure appropriate database alerts.	Craig, S. Database Performance Tuning and Optimization; Vendor Technical Guides.		
12	5. Maintain computerized database	Session 24: Database Maintenance Report and Integrated Maintenance	By the end of the session, trainees should be able to: 1. Analyse database maintenance	1. Provides maintenance findings and guides trainees in analysing them. 2. Provides	1. Analyse the database maintenance findings provided. 2. Evaluate database	Resources: Computers, DBMS, maintenance records, report templates,	Final Integrated Practical + Database Maintenance Report	Date: _____ Reflection: _____

		• Maintenance findings & optimization • Monitoring results • Backup/recovery records • Maintenance report	findings. 2. Evaluate database performance. 3. Apply corrective maintenance measures. 4. Organize maintenance records. 5. Prepare a database maintenance report. 6. Present maintenance findings professionally.	performance results and asks trainees to evaluate database performance. 3. Demonstrates corrective maintenance measures and guides trainees in applying them. 4. Demonstrates maintenance record organization and guides trainees in organizing records. 5. Demonstrates maintenance report preparation and guides trainees in preparing reports. 6. Demonstrates professional presentation techniques and guides trainees in presenting findings.	performance using the given results. 3. Apply appropriate corrective maintenance measures. 4. Organize database maintenance records. 5. Prepare a database maintenance report. 6. Present the maintenance findings professionally.	projector. References: TVET CDACC Assessment Standards; Mullin, M. Database Administration.		
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